

Lecture 2: Sept 12

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Contents: This lecture discusses some elementary properties of probability measures and introduces the notions of random variable and probability distribution of a random variable.

Textbook sections: II.1

2.1 Properties of probability measures

Two properties that are commonly taught in elementary courses in Probability and Statistics are the following

Monotonicity.

For any $A, B \in \mathcal{F}$ such that $A \subset B$, one has $\mathbb{P}[A] \leq \mathbb{P}[B]$

Inclusion/exclusion principle.

For any $n \geq 1$ and $A_1, \dots, A_n \in \mathcal{F}$ one has

$$\begin{aligned} \mathbb{P}\left[\bigcup_{i=1}^n A_i\right] &= \sum_{i=1}^n \mathbb{P}[A_i] - \sum_{i < j} \mathbb{P}[A_i \cap A_j] + \sum_{i < j < k} \mathbb{P}[A_i \cap A_j \cap A_k] \\ &\quad + \dots + (-1)^{n+1} \mathbb{P}[A_1 \cap \dots \cap A_n] \end{aligned}$$

For $n = 2$, it boils down to the familiar formula $\mathbb{P}[A_1 \cup A_2] = \mathbb{P}[A_1] + \mathbb{P}[A_2] - \mathbb{P}[A_1 \cap A_2]$. Further note that if $A_1, \dots, A_n$ were pairwise disjoint, then

$$\mathbb{P}\left[\bigcup_{i=1}^n A_i\right] = \sum_{i=1}^n \mathbb{P}[A_i]$$

which corresponds to finite additivity of $\mathbb{P}$.

We will examine in detail two additional properties. One establishes that a probability measure is continuous along monotonic sequences of events, while the other provides an upper bound for the probability of the union of a countable collection of events and is known as Boole's inequality.

Continuity of $\mathbb{P}$

Let $(A_n)_{n \geq 1}$ be an non-decreasing sequence of events in $\mathcal{F}$ such that $A_n \nearrow A \in \mathcal{F}$. Then $\mathbb{P}[A_n] \nearrow \mathbb{P}[A]$. If $(A_n)_{n \geq 1}$ were non-increasing and $A_n \searrow A$, then $\mathbb{P}[A_n] \searrow \mathbb{P}[A]$.

Proof. Since $(A_n)_{n \geq 1}$ is non-decreasing we may define a sequence of pairwise disjoint sets $(B_n)_{n \geq 1}$ as $B_1 = A_1$ and, for any $n \geq 2$, $B_n = A_n \setminus A_{n-1}$. Note that $B_i \cap B_j = \emptyset$ for any $i \neq j$,

$$A_n = \bigcup_{i=1}^n B_i \quad \text{and} \quad A = \bigcup_{i=1}^{\infty} B_i.$$

Hence, one has

$$\mathbb{P}[A] = \sum_{i=1}^{\infty} \mathbb{P}[B_i] = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{P}[B_i] = \lim_{n \rightarrow \infty} \mathbb{P}\left[\bigcup_{i=1}^n B_i\right] = \lim_{n \rightarrow \infty} \mathbb{P}[A_n]$$

where the third equality follows from finite additivity.

If $(A_n)_{n \geq 1}$ is non-increasing and such that $A_n \searrow A$, we may consider $(A_n^c)_{n \geq 1}$ since it is such that $A_n^c \nearrow A^c$. From the previous part of the proof, $\mathbb{P}[A_n^c] \nearrow \mathbb{P}[A^c]$ and

$$\mathbb{P}[A_n] = 1 - \mathbb{P}[A_n^c] \searrow 1 - \mathbb{P}[A^c] = \mathbb{P}[A].$$

This concludes the proof. $\square$

Boole's inequality

For any countable collection $A_1, A_2, \dots$ of sets in $\mathcal{F}$, one has

$$\mathbb{P}\left[\bigcup_{i=1}^{\infty} A_i\right] \leq \sum_{i=1}^{\infty} \mathbb{P}[A_i]$$

Proof. Define $C_1 = A_1$ and, for any $n \geq 2$, $C_n = A_1^c \cap \dots \cap A_{n-1}^c \cap A_n$. This clearly implies that $C_i \cap C_j = \emptyset$ for any $i \neq j$ and

$$\bigcup_{i=1}^n A_i = \bigcup_{i=1}^n C_i$$

which in turn entails

$$\mathbb{P}\left[\bigcup_{i=1}^n A_i\right] = \sum_{i=1}^n \mathbb{P}[C_i].$$

Since $C_i \subset A_i$ one has $\mathbb{P}(C_i) \leq \mathbb{P}(A_i)$ and, then

$$\mathbb{P}\left[\bigcup_{i=1}^n A_i\right] \leq \sum_{i=1}^n \mathbb{P}[A_i]$$

If we let $n \rightarrow \infty$, the right-hand side converges to $\sum_{i=1}^{\infty} \mathbb{P}[A_i]$ whereas the left-hand side converges, because of continuity, to $\mathbb{P}[\cup_{i=1}^{\infty} A_i]$ and the stated inequality follows. $\square$

2.2 Random variables

In what follows $(\Omega, \mathcal{F}, \mathbb{P})$ is the fundamental probability space

If $(E, \mathcal{E})$ is a measurable space, a function $X : \Omega \rightarrow E$ such that

$$X^{-1}(A) = \{\omega \in \Omega : X(\omega) \in A\} \in \mathcal{F},$$

for any $A \in \mathcal{E}$, is a *random variable*.

Remark. Based on a well-known result in measure theory, if $\mathcal{C}$ is a p-system that generates $\mathcal{E}$, i.e., $\sigma(\mathcal{C}) = \mathcal{E}$, then $X : \Omega \rightarrow E$ is a random variable if and only if

$$X^{-1}(A) \in \mathcal{F} \quad \text{for any } A \in \mathcal{C}.$$

Hence, one can confine oneself to checking that $X^{-1}(A) \in \mathcal{F}$ for any A in the smaller class $\mathcal{C} \subset \mathcal{E}$ rather than on the whole σ -algebra $\mathcal{E}$. This simplification is very useful, as we shall see in the next item.

Remark. While we gave a general definition that contemplates E -valued random variables, in this course we focus on cases where $E = \mathbb{R}$ (standard random variables, as in elementary probability courses) or $E = \mathbb{R}^d$ (random vectors) or $E = \mathbb{R}^\infty$ (random sequences or discrete time stochastic processes) or $E = \mathbb{R}^T$ (continuous-time stochastic processes, with T being some interval).

Examples

- (a) Let us consider $E = \mathbb{R}$ and $\mathcal{E} = \mathcal{B}(\mathbb{R})$. In Lecture 1 we have noted that the class of sets

$$\mathcal{C} = \{(-\infty, x] : x \in \mathbb{R}\}$$

is a p-system that generates $\mathcal{B}(\mathbb{R})$. Hence, X is a real-valued random variable if and only if $X^{-1}((-\infty, x]) = \{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$ for any $x \in \mathbb{R}$.

(b) Let $X(\omega) = k$ for any $\omega \in \Omega$ and for some constant $k \in \mathbb{R}$. Then

$$X^{-1}((-\infty, x]) = \begin{cases} \emptyset & \text{if } k > x \\ \Omega & \text{if } k \leq x \end{cases}$$

and, by definition, both $\emptyset$ and Ω belong to any σ -algebra $\mathcal{F}$ of subsets of Ω . Hence a constant function is measurable with respect to any σ -algebra on Ω and is, thus, a random variable.

(c) Let $A \subset \Omega$ and let X equal the indicator function $\mathbb{1}_A$ of set A , namely

$$X(\omega) = \mathbb{1}_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \notin A \end{cases}$$

Note that for any $x \in \mathbb{R}$

$$X^{-1}((-\infty, x]) = \begin{cases} \emptyset & \text{if } x < 0 \\ A^c & \text{if } 0 \leq x < 1 \\ \Omega & \text{if } x \geq 1. \end{cases}$$

Since $\emptyset, \Omega \in \mathcal{F}$ by definition of σ -algebra, $X = \mathbb{1}_A$ is a random variable if and only if $A \in \mathcal{F}$, as in this case also $A^c \in \mathcal{F}$.

(d) Let $\{A_1, \dots, A_N\}$ be a partition of Ω into $\mathcal{F}$ -sets, namely: (i) $A_i \in \mathcal{F}$ and $A_i \neq \emptyset$ for any i ; (ii) $A_i \cap A_j = \emptyset$ for any $i \neq j$; (iii) $A_1 \cup \dots \cup A_N = \Omega$. For a choice of $x_1 \neq x_2 \neq \dots \neq x_N$ in $\mathbb{R}$, set

$$X(\omega) = \sum_{i=1}^N x_i \mathbb{1}_{A_i}(\omega)$$

which implies that for any $\omega \in A_i$ one has $X(\omega) = x_i$. This is equivalent to $A_i = X^{-1}(\{x_i\}) = \{\omega \in \Omega : X(\omega) = x_i\}$. It can be seen that X is a random variable and this depends on having $A_i \in \mathcal{F}$ for any $i = 1, \dots, N$. Indeed, one has

$$X^{-1}((-\infty, x]) = \bigcup_{\{i: x_i \leq x\}} A_i$$

which equals: (i) $\emptyset$ if $\{i : x_i \leq x\} = \emptyset$; (ii) Ω if $\{i : x_i \leq x\} = \{1, \dots, N\}$ and (iii) finite union of A_i 's in all other cases.

A random variable taking on a finite number of values, namely

$$X(\omega) = \sum_{i=1}^N x_i \mathbb{1}_{A_i}(\omega)$$

for some $N \geq 1$ and $\{A_1, \dots, A_N\}$ a partition of Ω into $\mathcal{F}$ -sets, is said *simple*.

2.3 Distribution of a random variable

If X is a random variable defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and taking values in $(E, \mathcal{E})$, the probability distribution of X , say $\mathbb{P}_X$, is the image measure of $\mathbb{P}$ under X , namely

$$\mathbb{P}_X(A) = \mathbb{P}[X \in A] = \mathbb{P}[X^{-1}(A)] \quad \forall A \in \mathcal{E}.$$

This also motivates the notation $\mathbb{P}_X = \mathbb{P} \circ X^{-1}$. Note that:

✓ $\mathbb{P}$ is a probability measure on $(\Omega, \mathcal{F})$

✓ $\mathbb{P}_X$ is a probability measure on $(E, \mathcal{E})$

⇒ **Characterizing $\mathbb{P}_X$:** by definition, two random variables X and Y are equal in distribution if and only if $\mathbb{P}_X(A) = \mathbb{P}_Y(A)$ for any $A \in \mathcal{E}$. The notation $X \stackrel{d}{=} Y$ will be used in this case. Hence, checking whether two probability distributions coincide amounts to establishing that they are equal for any measurable set A and this may be difficult to ascertain. Luckily enough there is a result that considerably simplifies such a procedure.

Theorem. Let $\mathcal{C}$ be a p -system on E generating $\mathcal{E}$, namely $\sigma(\mathcal{C}) = \mathcal{E}$. If $\mathbb{P}$ and $\mathbb{Q}$ are probability measures on $(E, \mathcal{E})$ such that $\mathbb{P}(A) = \mathbb{Q}(A)$ for any $A \in \mathcal{C}$, then $\mathbb{P} = \mathbb{Q}$ on $\mathcal{E}$.

In view of this result, for example, we may say that $X \stackrel{d}{=} Y$ if and only if $\mathbb{P}_X(A) = \mathbb{P}_Y(A)$ for any A in the p -system $\mathcal{C}$, which is a smaller class than $\mathcal{E}$ and in several cases it is much easier to check this condition. An elementary example is with $E = \mathbb{R}$, then $\mathcal{C} = \{(-\infty, x] : x \in \mathbb{R}\}$ is a p -system generating $\mathcal{E} = \mathcal{B}(\mathbb{R})$. Hence $\mathbb{P}_X$ is identified by the function

$$x \mapsto F_X(x) = \mathbb{P}_X((-\infty, x]) = \mathbb{P}[X \leq x],$$

which is also known as the *distribution function* (or *cumulative distribution function*) of the random variable X . Hence, we have

$$X \stackrel{d}{=} Y \quad \Longleftrightarrow \quad F_X(x) = F_Y(x) \quad \text{for any } x \in \mathbb{R}$$

and this is much easier to check since it involves real functions.

The function $F_X : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the following properties

- (i) for any $x < y$ one has $F_X(x) \leq F_X(y)$, i.e. F_X is monotone non-decreasing
- (ii) $\lim_{x \rightarrow +\infty} F_X(x) = 1$ & $\lim_{x \rightarrow -\infty} F_X(x) = 0$
- (iii) F_X is continuous from the right, namely for any $x \in \mathbb{R}$ one has $\lim_{h \downarrow 0} F_X(x+h) = F_X(x)$.

$\Rightarrow$ **Probability measures on $E = \mathbb{R}$.** The previous theorem, that was used for characterizing $\mathbf{P}_X$, establishes a one-to-one correspondence between c.d.f.'s F_X and probability distributions $\mathbf{P}_X$. This provides a means for constructing probability measures on the real line. First, we introduce the following definition.

Any function $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfying properties (i)–(iii) above, it is known as *distribution function*.

As a consequence of the previous theorem, one can show that that a distribution function F uniquely identify a probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$?

Theorem

Suppose $F : \mathbb{R} \rightarrow \mathbb{R}$ is a distribution function. Then there exists a unique probability measure $\mathbf{P}$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ admitting F as its cumulative distribution function, namely

$$\mathbf{P}((-\infty, x]) = F(x) \quad \forall x \in \mathbb{R}.$$

Hence, a probability measure on $\mathbb{R}$ is uniquely identified by a distribution function F .

$\Rightarrow$ **Random variables with a given distribution function.** It has been shown that given a distribution function F , there exists a unique probability measure $\mathbf{P}$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $\mathbf{P}((-\infty, x]) = F(x)$ for any x in $\mathbb{R}$. One may now wonder whether there exists a probability space $(\Omega, \mathcal{F}, \mathbf{P})$ and a random variable X , defined on it, such that F is the distribution function of X . The answer is positive and straightforwardly follows from taking $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}[0, 1]$ and $\mathbf{P} = \text{Leb}$ and

$$X(\omega) = \inf\{z \in \mathbb{R} : F(z) \geq \omega\}.$$

Hence

$$\{\omega \in [0, 1] : X(\omega) \leq x\} = \{\omega \in [0, 1] : \omega \leq F(x)\}$$

and this implies that for any $x \in \mathbb{R}$

$$F_X(x) = \mathbf{P}[X \leq x] = \text{Leb}(\{\omega \in [0, 1] : \omega \leq F(x)\}) = F(x).$$